

Vincent Zhou

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EDUCATION

University of Waterloo
Bachelor of Applied Science, Computer Engineering, Honours Co-op

Sept 2025 – June 2030

EXPERIENCE

Waterloo Aerial Robotics Group (WARG)

Autonomy Project Manager

Sept 2026 – Present

- Lead a **30-member** autonomy subteam, organizing contributors into parallel project groups, scoping the flight-software backlog into issues sized for onboarding, and tracking them through to flight readiness for the 2027 AEAC competition.
- Run code review for the autonomy stack and set its architectural direction, defining the ROS 2 node and message boundaries between the perception, GNC, and obstacle-avoidance packages so work split across groups integrates without rework.

Software Engineering Intern

May 2026 – Sept 2026

- Independently designed, built, and flight-tested a **visual teach-and-repeat** precision-landing system 🌀 (ROS 2, OpenCV, MAVROS) that records an altitude-keyed map of **ORB** features during takeoff, then on descent matches live frames against the nearest teach frame using Lowe-ratio-filtered kNN matching, metric back-projection with roll/pitch compensation, and RANSAC affine estimation to guide the drone back onto its launch point, landing within **15 cm in flight test** and **2 cm in simulation**.
- Validated the descent controller in isolation against an **AprilTag** fiducial baseline 🌀 first, confirming landings on the marker before substituting feature-based localization, so control and perception failures could be diagnosed separately.
- Hardened the pipeline against field failure modes: an altitude-proportional alignment cone that tapers descent authority with match quality, a stale-vision recovery state machine that climbs to widen the camera footprint and re-acquire, teach-frame keypoint gating, and inlier-ratio and scale validation to reject degenerate transforms.
- Deployed on an **NVIDIA Jetson** with a **CUDA** brute-force matcher (CPU fallback) inside a **10 Hz** control loop; wrote the Arducam V4L2 driver, systemd service, and **Bash** provisioning scripts, and handed off touchdown to **ArduPilot**'s LAND mode with debounced ground confirmation.
- Architected a full-stack, real-time **UAV geolocation platform** (React/TypeScript frontend, Flask REST API) for the 2026 AEAC Firefighting Competition, enabling ground operators to capture, annotate, and geolocate targets spotted mid-flight using a pinhole camera model and ray-plane intersection against live telemetry to localize targets in 3D to **<0.5 m**.

PROJECTS

SITL-Plus 🌀 | UAV Software-in-the-Loop Simulator, WARG

- Built a from-scratch simulator using **PyBullet** rigid-body physics to model a quadcopter's flight dynamics, bridged to **ArduPilot**'s SITL binary over a **UDP** JSON protocol for high-rate physics/flight-controller state exchange.
- Simulated onboard sensors: downward and forward RGB-D cameras via PyBullet's OpenGL renderer and a ray-cast rangefinder/LiDAR, streamed to **Rerun** over **gRPC** for live 3D visualization, enabling integration testing of the vision and precision-landing pipelines without physical hardware.
- Containerized the stack (physics engine + ArduPilot SITL/MAVProxy) with **Docker Compose**, covered the sensor and dynamics models with a **pytest** suite, and used the simulator to validate and tune the precision-landing pipeline to **2 cm** before committing to flight tests.

CNN Feature Matching for Precision Landing | extends the WARG landing pipeline

Sept 2026 – Present

- Implementing a lightweight 2D convolutional detector-descriptor network (**XFeat**) with parallel keypoint-detection and 64-dimensional dense-descriptor heads, targeting the viewpoint and illumination changes where hand-engineered **ORB** degrades, such as the scale gap between a teach frame and a much lower repeat frame, or exposure and contrast shifts between takeoff and landing.
- Training on a self-captured aerial dataset with ground-truth correspondences generated from known homographies and photometric augmentation, then benchmarking match precision, **RANSAC** inlier ratio, and landing error against the **ORB** baseline over identical replayed trajectories, scored against the **Jetson** 10 Hz inference budget.

Classly 🌀 | Classroom Management Platform: React 19, Node.js/Express 5, TypeScript

- Built a full-stack school management platform with a **Postgres/Drizzle** ORM schema, role-based access control (admin/teacher/student), **Better-Auth** session authentication (email/password, Google/GitHub OAuth), and invite-based onboarding over cryptographically secure tokens.
- Migrated the backend off a long-running **Render** container onto **Cloudflare Workers** with serverless Postgres over WebSockets, scoping connection-pool lifetime to the request to fix pools leaking across transactions, and added a cron-triggered session-cleanup job.

Neural Network From Scratch 🌀 | C++, Eigen

- Implemented a multilayer perceptron from scratch with no ML framework, deriving forward and backpropagation by hand, with mini-batch SGD and early stopping that checkpoints the best weights on validation plateaus, reaching **95.4%** test accuracy on MNIST.
- Built a **raylib** GUI for real-time handwritten-digit drawing with MNIST-style preprocessing (centering, normalization) and multithreaded live inference, over a custom big-endian IDX file parser.

SKILLS

- **Languages:** Python, C++, TypeScript/JavaScript, Java, SQL, Bash, Lua, Verilog
- **Robotics & Vision:** ROS 2, ArduPilot/MAVROS, MAVLink, OpenCV, CUDA, PyBullet, Rerun, NumPy, Eigen, NVIDIA Jetson, Raspberry Pi
- **Backend & Data:** FastAPI, Flask, Node.js/Express, REST, PostgreSQL, Drizzle ORM, Neon, Pydantic, Zod
- **Infrastructure & Testing:** Docker, Cloudflare Workers/Pages, Wrangler, Git/GitHub, GitHub Actions CI, Linux/systemd, pytest, Vitest, Quartus/FPGA